VOLTAGE HEADROOM / USABLE BATTERY

Chiralight standard-motor

More usable battery. More room for payload.

More of your battery is useful when the motor drive retains thrust at lower voltage. This simulated 3 kg quad uses 10-inch props, a 640KV motor and a 6S 4 Ah pack, while retaining a **3:1 thrust-to-weight ratio (TWR)** maneuver reserve for gust recovery and obstacle avoidance.

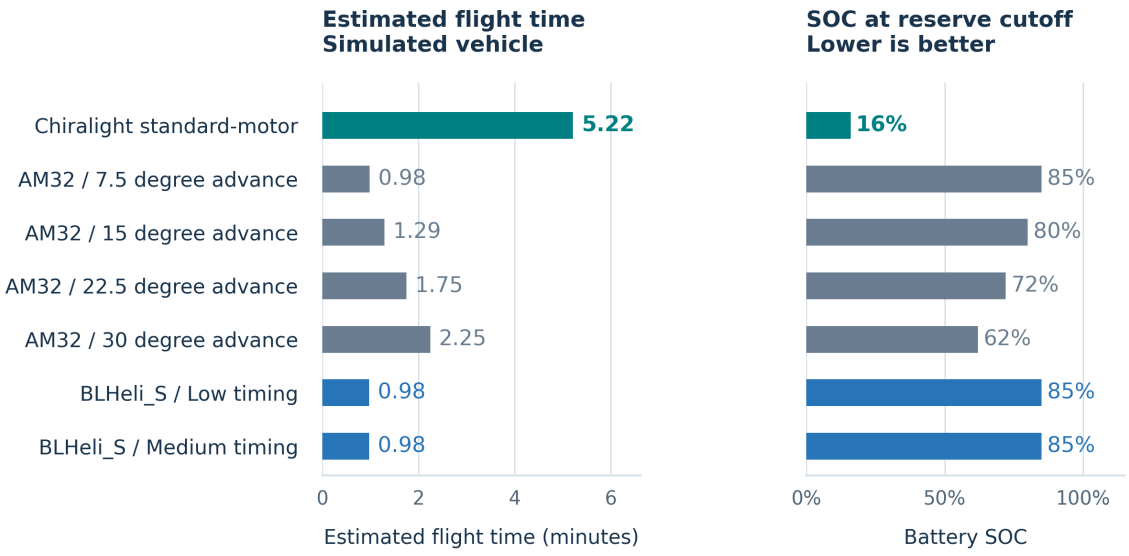
Up to 100%
or greater
More modeled flight time
Simulation results

Gain margin
More voltage headroom
for your chosen motor

+174 g
Potential payload by
choosing 6S over 8S

3:1 TWR MANEUVER RESERVE FOR GUST RECOVERY AND OBSTACLE AVOIDANCE

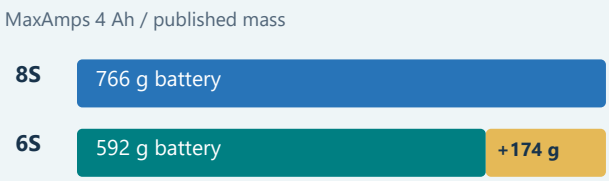
640KV / 6S | 3:1 TWR maneuver reserve



Estimated flight times for a simulated 3 kg vehicle at a 1.5:1 TWR flight load, with 3:1 TWR maneuver reserve.

6S FOR PAYLOAD. 8S FOR ADDITIONAL VOLTAGE RESERVE.

The smaller 4 Ah pack releases **174 g for payload** at the same 3 kg all-up mass. The larger pack stores more energy and gives the drive more voltage headroom.



Similar energy, less power: MaxAmps 8S 2.8 Ah vs 6S 4 Ah (83 vs 89 Wh), but **38% lower peak and 27% lower continuous rated power**. Matching Wh does not guarantee sustained maneuver reserve.

Reserve-cutoff rule: the selected SOC must retain the required thrust and at least 3.0 V/cell under the maneuver load. All ESCs receive adequate DC-link buffering; sustained battery resistance loss remains. Flight-time estimates use the simulated 1.5:1 TWR load; headline versus AM32 at 30 degree advance.

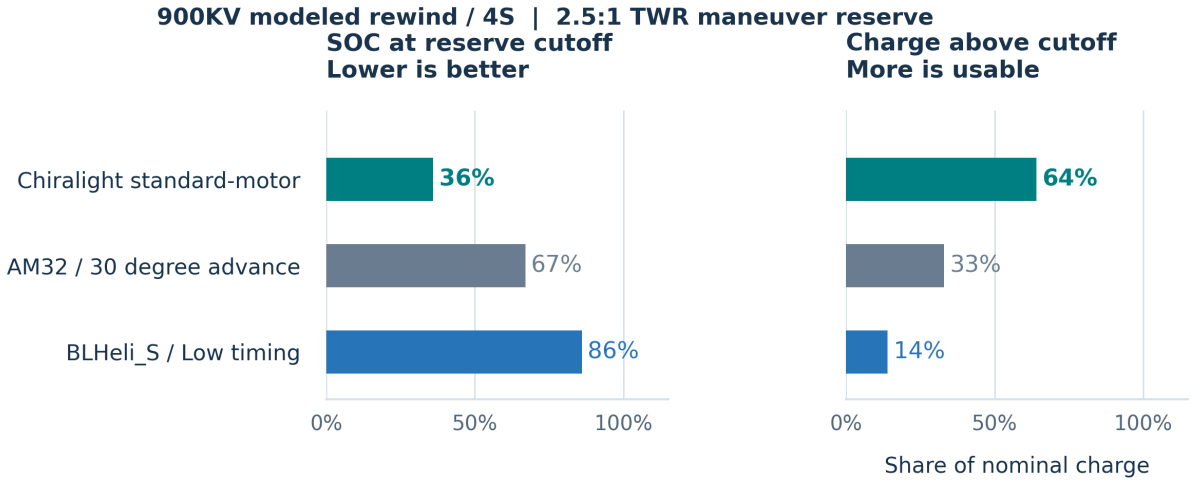
Verifiable simulation results. Contact Chiralight Systems for details.
Evaluate for your aircraft. Exact performance may vary.

CONFIGURATION TRADEOFFS / REPRODUCIBLE RESULTS

Match the drive to the mission.

Modeled 900KV rewind / 4S: lower battery mass

The lower-S, higher-KV option trades maneuver reserve for battery mass: 2.5:1 TWR, rather than 3:1 TWR. Chiralight standard-motor again retains thrust deeper into discharge, opening another route to a smaller battery. Winding, pack and maneuver reserve remain aircraft design choices.



Each competitor uses its lowest tested, source-checked reserve cutoff in this configuration.

4S FOR PAYLOAD. 6S FOR ADDITIONAL VOLTAGE RESERVE.

The smaller 4 Ah pack releases **182 g for payload** at the same 3 kg all-up mass. The larger pack stores more energy and gives the drive more voltage headroom.



FLIGHT POWER AND MANEUVER RESERVE ARE DIFFERENT REQUIREMENTS

1.5:1 TWR flight load: 1,125 gf per rotor supplies the mission-power estimate.
3:1 TWR maneuver reserve: 2,250 gf per rotor is the 640KV maneuver gate.
2.5:1 TWR reduced reserve: 1,875 gf per rotor is the explicitly reduced 900KV gate.
All four motors share the battery. Each reserve check includes the higher current and loaded-voltage sag.

Built from running code. The bench runs compiled Chiralight standard-motor, AM32 and official BLHeli_S code against the same nonlinear B/H motor plant and EMAX-table-informed propeller load. The reserve cutoffs come from 298 native runs, with source-update and buffering checks. The model uses buffered 4 Ah packs, 5 milliohms per series cell, a synthetic LiPo OCV/SOC curve, four motors and 10 W auxiliaries. FOC uses 1 A injection, MTPA and overmodulation with dedicated field weakening disabled.

Published battery references: MaxAmps LiPo 4 Ah 4S, 410 g / 6S, 592 g / 8S, 766 g. Contact Chiralight Systems for source traces, controller settings and reproduction instructions.